# CS182: Introduction to Machine Learning, Fall 2025
# Homework 2

(Due on Friday, Nov. 14 at 11:59 pm (CST))

Notice:

- Please submit your assignments via Gradescope. The entry code is 428YBY.
- Please make sure you select your answer to the corresponding question when submitting your assignments.
- Each person has a total of five days to be late without penalty for all the assignments. Each late delivery less than one day will be counted as one day.

1. [20 points] [*Parameter Estimation*] A 2-axis force sensor, mounted on a robot fingertip, measures an applied force $\mathbf{F} = (F_x, F_y)$. Due to manufacturing tolerances and mechanical crosstalk, the sensor's voltage output $V_x$ is linearly dependent on both $F_x$ and $F_y$.

   We use a Multivariate Linear Regression Model to calibrate the $V_x$ channel. This model includes a bias term $b$ and two gain terms, $m_{11}$ and $m_{12}$:

   $$V_x = m_{11}F_x + m_{12}F_y + b$$

   We assume the sensor's measurement noise $\epsilon$ is Gaussian, $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Therefore, for the $i$-th measurement, the observation model is:

   $$V_x^i = (m_{11}F_x^i + m_{12}F_y^i + b) + \epsilon_i$$

   The parameter vector we want to estimate is $\mathbf{w} = \begin{bmatrix} m_{11} \\ m_{12} \\ b \end{bmatrix}$.

   (a) Given a set of $N$ i.i.d. calibration data points $\{X\} = \{(F_x^i, F_y^i), V_x^i\}_{i=1}^N$. Write the Log-Likelihood Function $\mathcal{L}(\mathbf{w}|\{X\})$ for the parameters $\mathbf{w}$. [4 points]

   (b) We know that maximizing $\mathcal{L}(\mathbf{w}|\{X\})$ is equivalent to minimizing the Sum of Squared Errors (SSE) $E(\mathbf{w}) = \sum_{i=1}^N (V_x^i - (m_{11}F_x^i + m_{12}F_y^i + b))^2$.[8 points]

   - Write the SSE $E(\mathbf{w})$ in matrix-vector form: $E(\mathbf{w}) = \|\mathbf{r} - \mathbf{X}\mathbf{w}\|^2$. Clearly define the design matrix $\mathbf{X}$, the parameter vector $\mathbf{w}$, and the observation vector $\mathbf{r}$ based on the model.
   - By computing the gradient of $E(\mathbf{w})$ with respect to the vector $\mathbf{w}$ (i.e., $\nabla_{\mathbf{w}} E(\mathbf{w})$) and setting it to zero, derive the Normal Equations: $(\mathbf{X}^T\mathbf{X})\hat{\mathbf{w}} = \mathbf{X}^T\mathbf{r}$,

   (c) During an experiment, $N = 4$ known forces were applied, and the corresponding $V_x$ voltages were measured.[8 points]

| Measurement $i$ | Applied Force $(F_x, F_y)$ (N) | Measured Voltage $V_x$ (V) |
|---|---|---|
| 1 | (1, 0) | 2.1 |
| 2 | (0, 1) | 0.9 |
| 3 | (1, 1) | 3.0 |
| 4 | (2, 0) | 4.1 |

   - Using the data provided in the table, first compute the numerical values for $\mathbf{X}^T\mathbf{X}$ and $\mathbf{X}^T\mathbf{r}$. Then, solve the system to find the optimal parameter estimates $\hat{\mathbf{w}} = (\hat{m}_{11}, \hat{m}_{12}, \hat{b})^T$.
   - A unique least-squares solution $\hat{\mathbf{w}}$ only exists if the applied calibration forces satisfy a specific condition. From an experimental design perspective, what **physical requirement** must the set of $N = 4$ applied forces $\{(F_x^i, F_y^i)\}$ meet to ensure that the parameters $(m_{11}, m_{12}, b)$ can be uniquely determined? Briefly explain why.

2. [20 points] [*Parameter Estimation*] Assume that the prior distribution of $\theta$ follows a Gaussian distribution:

$$p(\theta) = \mathcal{N}(0, \sigma^2),$$

where $\theta$ is an unknown parameter and $\sigma$ is known. The conditional distribution of $x$ given $\theta$ is

$$p(x \mid \theta) = \mathcal{N}(\theta, 1).$$

Please provide detailed derivations for the following:

(a) Derive the marginal distribution $p(x)$. (Hint: $\int_{-\infty}^{\infty} e^{-ax^2+bx+c}dx = \sqrt{\frac{\pi}{a}}e^{\frac{b^2}{4a}+c}$.) [6 points]

(b) Compute the posterior expectation $\mathbb{E}(\theta \mid x)$. [6 points]

(c) Suppose the prior $p(\theta)$ is unknown, and we only know that $p(x \mid \theta) = \mathcal{N}(\theta, \sigma^2)$. Find the posterior expectation $\mathbb{E}(\theta \mid x)$. [8 points]

3. [20 points] [*Parameter Estimation*]

In class, we have learned that the MMSE is achieved when using the conditional expectation $\mathrm{E}[X|\mathbf{Y}]$, i.e., $\arg\min_{\hat{X}} \mathrm{E}[|X-\hat{X}(\mathbf{Y})|^2] = E[X|\mathbf{Y}]$, where $X$ is some unknown random variable and $\mathbf{Y}$ is the observation. However, $\mathrm{E}[X|\mathbf{Y}]$ can be intractable to compute. Now, let's focus on linear estimation and minimize MSE over all linear estimators.

**Definition 1** (LLSE)**.**

$$\mathbb{L}[X|\mathbf{Y}] = \arg \min_{\text{linear } \hat{X}} \mathrm{E}[|X-\hat{X}(\mathbf{Y})|^2]$$

where linear estimators $\hat{X}(\mathbf{Y})$ are of the form

$$\hat{X}(\mathbf{Y}) = a + \sum_{i=1}^{n} b_i Y_i$$

for $a, b_i \in \mathbb{R}$ where $\mathbf{Y} = (Y_1, \cdots, Y_n)$ is a vector of observations.

(a) Define

$$\Sigma_{\mathbf{XY}} = \mathrm{E}[(X-\mu_x)(\mathbf{Y}-\mu_{\mathbf{Y}})^T] \qquad \text{(cross-covariance matrix)}$$
$$\Sigma_{\mathbf{Y}} = \mathrm{E}[(\mathbf{Y}-\mu_{\mathbf{Y}})(\mathbf{Y}-\mu_{\mathbf{Y}})^T] \qquad \text{(covariance matrix)}$$

where $\mu_X = \mathrm{E}(X), \mu_Y = \mathrm{E}[Y] = [\mathrm{E}[Y_1], \cdots, \mathrm{E}[Y_n]]$
Show that

$$\mathbb{L}[X|\mathbf{Y}] = \mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}(\mathbf{Y}-\mu_{\mathbf{Y}}) \tag{1}$$

[5 points]

(b) Show that a linear estimator $\hat{X}(\mathbf{Y})$ achieves minimum mean square error(over all linear estimators) if and only if

$$\mathrm{E}[\hat{X}(\mathbf{Y}) - X] = 0 \tag{2}$$
$$\mathrm{E}[(\hat{X}(\mathbf{Y}) - X)Y_i] = 0 \tag{3}$$

This is called the **orthogonality principle** since it uniquely characterizes $\mathbb{L}[X|\mathbf{Y}]$.
[10 points]

(c) In the real world, data is observed sequentially. How do we efficiently update our LLS estimate on the arrival of new observations? For motivation, start with a simple setting. w.l.o.g., assume $\mathrm{E}[X] = 0$, Suppose observations $Y_1, Y_2, \cdots$ are orthogonal and uncorrelated, i.e., $\langle Y_i, Y_j \rangle = \mathrm{E}[Y_i Y_j] = \mathrm{Cov}(Y_i, Y_j) = 0$ for $i \neq j$ to each other.
For convenience, define

$$\mathbf{Y}^n = (Y_1, \cdots, Y_n)$$
$$\mu_i = \mathrm{E}[Y_i]$$

Show that

$$\mathbb{L}[X|\mathbf{Y}^{n+1}] = \mathbb{L}[X|\mathbf{Y}^n] + \mathbb{L}[X|Y_{n+1}] \tag{4}$$
$$= \mathbb{L}[X|\mathbf{Y}^n] + \frac{\mathrm{Cov}(X, Y_{n+1})}{\mathrm{Var}(Y_{n+1})}(Y_{n+1} - \mu_{n+1}) \tag{5}$$

[5 points]
(Hint: Just check the orthogonality principle!)

4. [20 points] [*Linear Discriminant Analysis*] We have a two-class dataset with label $\mathcal{Y} = \{-1, +1\}$ and the feature vector $\boldsymbol{x} \in \mathbb{R}^2$. We find the means of these two classes are $\hat{\boldsymbol{\mu}}_{-1} = (-1, -1)^\top$ and $\hat{\boldsymbol{\mu}}_{+1} = (1, 1)^\top$, respectively. And the estimated prior probabilities for these two class are equal (i.e., $\hat{\pi}_{+1} = \hat{\pi}_{-1} = \frac{1}{2}$). Some additional explanations on question 4. The Fisher's linear discriminant is a supervised dimensionality reduction method. The terms Fisher's linear discriminant and LDA are often interchangeably. The Fisher's linear discriminant maximizes the ratio between $\boldsymbol{S}_B$ and $\boldsymbol{S}_W$, where $\boldsymbol{S}_B$ denotes the between-class scatter matrix and $\boldsymbol{S}_W$ denotes the within-class scatter matrix, and this method is based on the so-called Fisher's linear discriminant rule. However, there is another type of LDA, which is based on Bayes discriminant rule, is presented to you in this question. By assuming $p(\boldsymbol{x}|C_i) \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$ (i.e., homoscedasticity), this type of LDA assigns $\boldsymbol{x}$ to the group that maximizes $\pi_i p(\boldsymbol{x}|C_i)$, where $\pi_i$ represents the prior probability of each classification $C_i$, and $p(\boldsymbol{x}|C_i)$ represent the $i$-th density. In question 4(a), we want you to use this type of LDA based on Bayes discriminant rule. In question 4(b), without homoscedasticity, the problem converts to quadratic discriminant analysis (QDA). You can get more detailed information through: `https://harvard-iacs.github.io/2018-CS109A/a-sections/a-section-6/presentation/a_section6.pdf`

   (a) Assuming the covariance martix can be estimated as

   $$\hat{\boldsymbol{\Sigma}} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

   with $0 \leq \rho < 1$ when applying LDA. Find the decision boundary as a function of $\rho$. [7 points]

   (b) Now suppose that we model each class with its own covariance matrix. The covariance matrix of class $+1$ is

   $$\hat{\boldsymbol{\Sigma}}_{+1} = \begin{pmatrix} 1/5 & 0 \\ 0 & 5 \end{pmatrix}.$$

   And the covariance matrix of class $-1$ is

   $$\hat{\boldsymbol{\Sigma}}_{-1} = \begin{pmatrix} 5 & 0 \\ 0 & 1/5 \end{pmatrix}.$$

   Describe the decision rule. [8 points]

   (c) When $p(\boldsymbol{x}|C_i) \sim \mathcal{N}\left(\boldsymbol{\mu}_i, \hat{\boldsymbol{\Sigma}}\right)$ (i.e., homoscedasticity), using Fisher's linear discriminant, prove that the vector $\mathbf{w}$ on which the data $\boldsymbol{x}$ are projected is $\mathbf{w} = \hat{\boldsymbol{\Sigma}}^{-1}(\boldsymbol{\mu}_{+1} - \boldsymbol{\mu}_{-1})$. [5 points]

5. [20 points] [*Logistic Discrimination*] Use a quadratic discriminant classifier to classify the $K$ classes $C_1, \ldots, C_K$. Assume that the class-conditional distributions are multivariate normal:

$$p(\mathbf{x} \mid C_i) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

where $\boldsymbol{\mu}_i$ is the mean vector and $\boldsymbol{\Sigma}_i$ is the covariance matrix for class $C_i$.

The quadratic discriminant function is given by:

$$\log \frac{p(\mathbf{x} \mid C_i)}{p(\mathbf{x} \mid C_K)} = \mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0} \tag{6}$$

where $\mathbf{W}_i \in \mathbb{R}^{d \times d}$ is a symmetric matrix related to the covariance matrices, $\mathbf{w}_i \in \mathbb{R}^d$ is a vector, and $w_{i0} \in \mathbb{R}$ is a scalar.

(a) Show the relationship between the parameters $(\mathbf{W}_i, \mathbf{w}_i, w_{i0})$ and the Gaussian parameters $(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$. [5 points]

(b) Using the quadratic discriminants in equation (6), derive the expression for the posterior probability $P(C_i \mid \mathbf{x})$ for each class $i = 1, \ldots, K$. Your answer should involve a softmax function. [5 points]

(c) Show that if all classes share the same covariance matrix, i.e., $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$ for all $i$, then the decision boundaries become linear. Also, find the expressions for the parameters $\mathbf{W}_i$, $\mathbf{w}_i$, and $w_{i0}$. [5 points]

(d) Discuss the advantages and disadvantages of quadratic discriminant compared to linear discriminant. Consider aspects such as model flexibility, number of parameters, and risk of overfitting. [5 points]